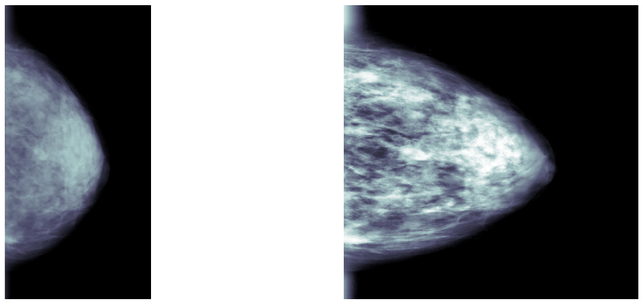
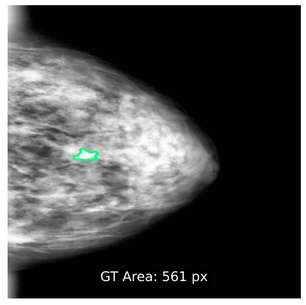
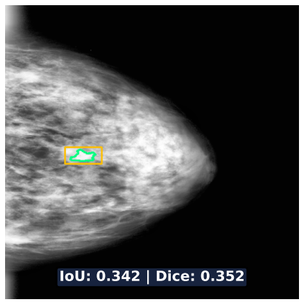
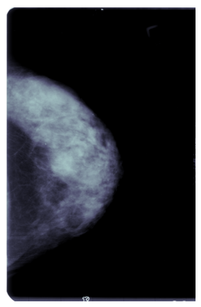
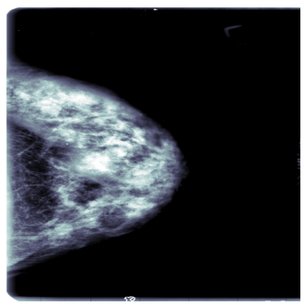
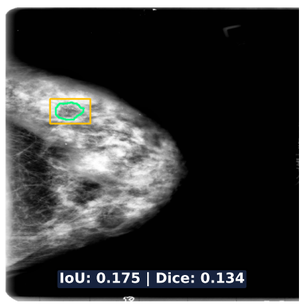
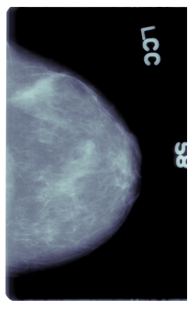
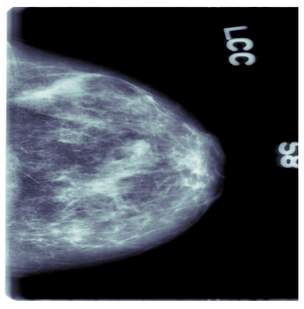
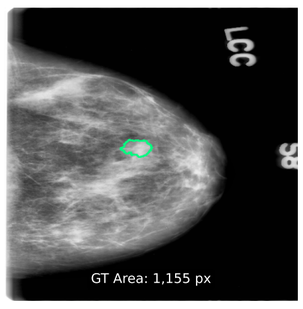
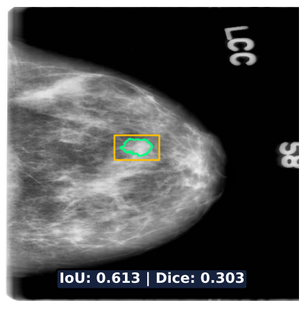
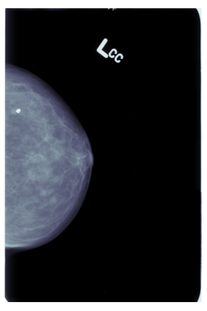
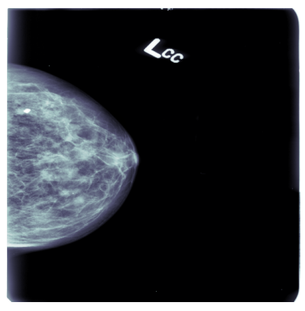
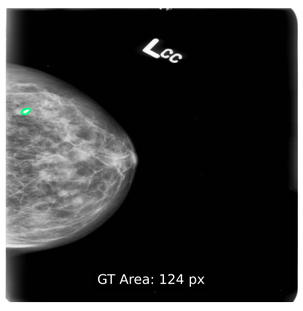
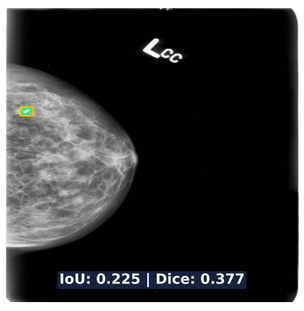
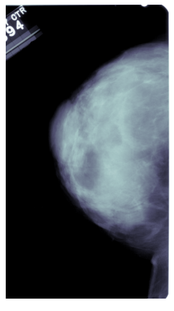
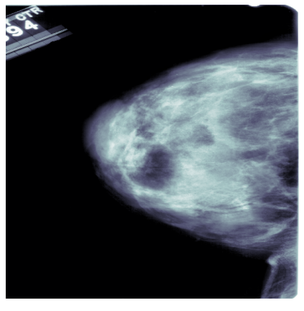
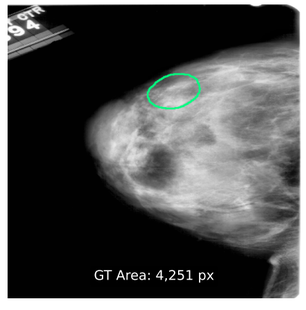
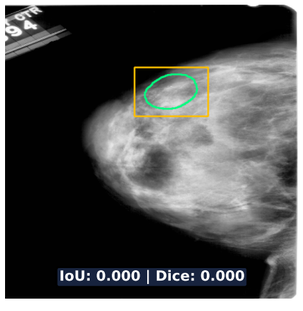
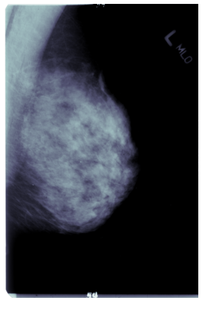
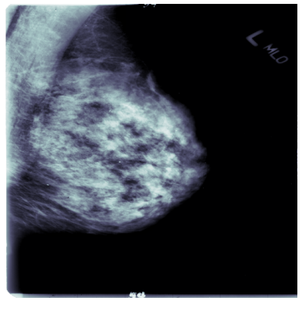
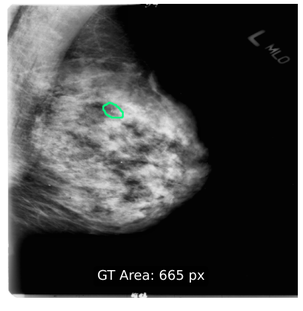
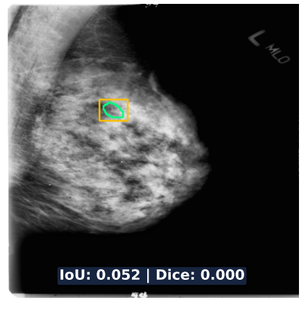
(a) Raw Mammogram	(b) Wiener + CLAHE Enhanced	(c) Ground-Truth Lesion Mask	(d) RoI Candidate Detection	(e) Diagnostic Decision Profile
CASE-TP1 (Right CC) **True Positive**					DIAGNOSTIC OUTCOME: CORRECT CLASSIFICATION Case Identifier: CASE-TP1 (Right CC) Ground Truth: MALIGNANT (MASS) Model Decision: MALIGNANT (Threshold tau=0.50) Predicted Risk: P(Malignant) = 0.749 P(Benign) = 0.252 Localization: IoU = 0.342 | Dice = 0.352 Observation: Irregular high-density mass with radiating architectural distortion. Accurately localized by U-KAN and assigned high malignant risk.
CASE-TP2 (Left CC) **True Positive**					DIAGNOSTIC OUTCOME: CORRECT CLASSIFICATION Case Identifier: CASE-TP2 (Left CC) Ground Truth: MALIGNANT (CALCIFICATION) Model Decision: MALIGNANT (Threshold tau=0.50) Predicted Risk: P(Malignant) = 0.979 P(Benign) = 0.021 Localization: IoU = 0.175 | Dice = 0.134 Observation: Fine pleomorphic microcalcifications in clustered ductal pattern. Classifier correctly recognized microcalcification patch signature.
CASE-TN1 (Left CC) **True Negative**					DIAGNOSTIC OUTCOME: CORRECT CLASSIFICATION Case Identifier: CASE-TN1 (Left CC) Ground Truth: BENIGN (MASS) Model Decision: BENIGN (Threshold tau=0.50) Predicted Risk: P(Malignant) = 0.029 P(Benign) = 0.971 Localization: IoU = 0.613 | Dice = 0.303 Observation: Circumscribed oval density consistent with benign fibroadenoma. High localization overlap (IoU=0.61) with low malignant risk (P=0.03).
CASE-TN2 (Left CC) **True Negative**					DIAGNOSTIC OUTCOME: CORRECT CLASSIFICATION Case Identifier: CASE-TN2 (Left CC) Ground Truth: BENIGN_WITHOUT_CALLBACK (CALCIFICATION) Model Decision: BENIGN (Threshold tau=0.50) Predicted Risk: P(Malignant) = 0.026 P(Benign) = 0.974 Localization: IoU = 0.225 | Dice = 0.377 Observation: Scattered punctate calcifications. Correctly ruled out with minimal malignant posterior probability (P=0.03).
CASE-FP1 (Left CC) **False Positive Error**					DIAGNOSTIC OUTCOME: DIAGNOSTIC ERROR Case Identifier: CASE-FP1 (Left CC) Ground Truth: BENIGN (CALCIFICATION) Model Decision: MALIGNANT (Threshold tau=0.50) Predicted Risk: P(Malignant) = 0.585 P(Benign) = 0.415 Localization: IoU = 0.000 | Dice = 0.000 Observation: Dense fibro-glandular parenchyma background caused over-estimation of malignancy risk (P=0.58 vs threshold 0.50).
CASE-FN1 (Left MLO) **False Negative Error**					DIAGNOSTIC OUTCOME: DIAGNOSTIC ERROR Case Identifier: CASE-FN1 (Left MLO) Ground Truth: MALIGNANT (CALCIFICATION) Model Decision: BENIGN (Threshold tau=0.50) Predicted Risk: P(Malignant) = 0.215 P(Benign) = 0.785 Localization: IoU = 0.052 | Dice = 0.000 Observation: Pectoral overlap and projection distortion obscured calcification clusters on MLO view, leading to low risk prediction (P=0.22).